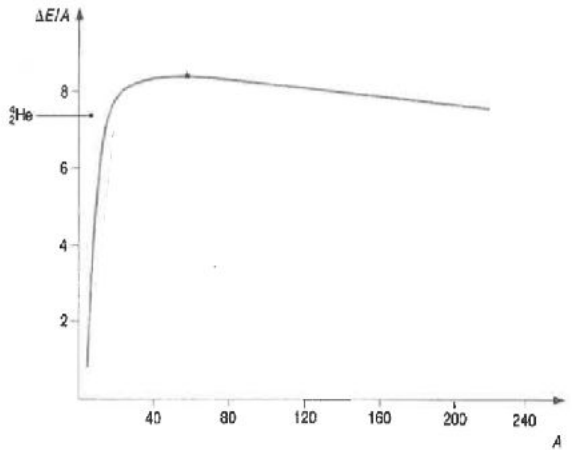
8(a)(i)	mass defect $\Delta m = 2 \times 1.007825u + 2 \times 1.008665u - 4.002604u = 0.030376u$	M1
	binding energy per nucleon, $E = \Delta mc^2 = \frac{0.030376 \times 1.66 \times 10^{-27} \times (3.00 \times 10^8)^2}{1.60 \times 10^{-19} \times 10^6} \times \frac{1}{4} = 7.09$ MeV	A1
8(a)(ii)	 shape of graph. Upward sloping portion is about ¼ of graph with peak between *A* = 40 and *A* = 80. Downward sloping portion is about ¼ of upper portion of graph.	B1

8(a)(iii)	Product has higher binding energy per nucleon / more stable.	B1
8(a)(iv)	mass loss $\Delta m = 2.014102u + 1.007825u - 3.01603u = 0.005897u$ energy released $E = \Delta mc^2 = \frac{0.005897\times1.66\times10^{-27}\times(3.00\times10^{8})^2}{1.60\times10^{-19}\times10^{6}} = 5.51$ MeV	B1
8(b)	The decay of each particular parent nucleus releases the same amount of energy. and Some kinetic energy must have been carried away by another particle. Also accept: sum of the positrons' energy and neutrino's energy is constant for the decay of a particular nuclide	B1
	Positrons having varying kinetic energy implies have varying momentum (since $E = \frac{p^2}{2m}$) If there were only 2 particles, there would be a fixed momentum. (by principle of conservation of momentum)	B1
8(c)	Electric potential energy = $\frac{1}{4\pi\varepsilon_0}\frac{q_1q_2}{r} = 8.99\times10^{9}\frac{(1.6\times10^{-19})^2}{1.0\times10^{-14}} = 2.30144\times10^{-14}$ J	M1
	Using $E = \frac{3}{2}kT$, $2.30144\times10^{-14} = \frac{3}{2}\times1.38\times10^{-23}T$, giving $T = 1\times10^{9}$ K Also accept if students consider two proton approaching each other, hence uses $2\times\frac{3}{2}kT =$ electric potential energy, resulting in 0.6×10^{9} K	A1
8(d)	6 – 2 = 4 protons	B1
8 (e)(i)	$\frac{0.1\times2.0\times10^{30}}{1.67\times10^{-27}}$ or $\frac{0.1\times2.0\times10^{30}}{1.007825\times1.66\times10^{-27}} = 1.1976\times10^{56}$	B1
(ii)	Energy = power × time $1.1976\times10^{56}\times6.68\times10^{6}\times1.6\times10^{-19} = 3.8\times10^{26}t$ giving $t = 1.07\times10^{10}$ years	B1